
SWS3009 – Part B

Robotics / TeleOperation

Section 1: Introduction

Dr. Boyd Anderson

I am a Kiwi (New Zealander) and did my PhD at NUS. I liked it so much here that I decided I didn't want to leave!



My research interests are in *embedded systems*, *wearable sensors*, and *gait analysis* (analysing the way you walk or run). I am also working on *wearable sensors for sports* (such as fencing and sprinting).

Contact me at: boyd@comp.nus.edu.sg

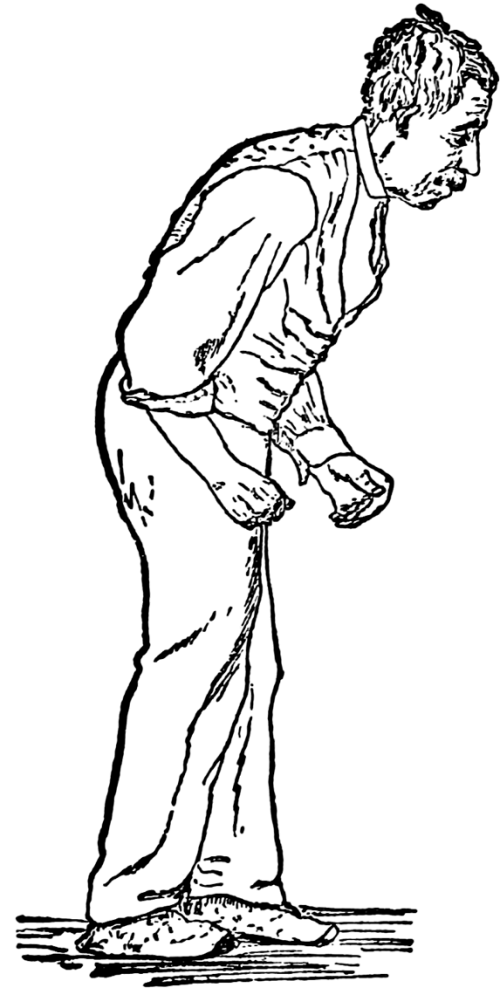
My office: **COM3-02-46**

Research Areas

Gait Analysis is the study and quantification of human locomotion.

Why are we interested in Gait Analysis?

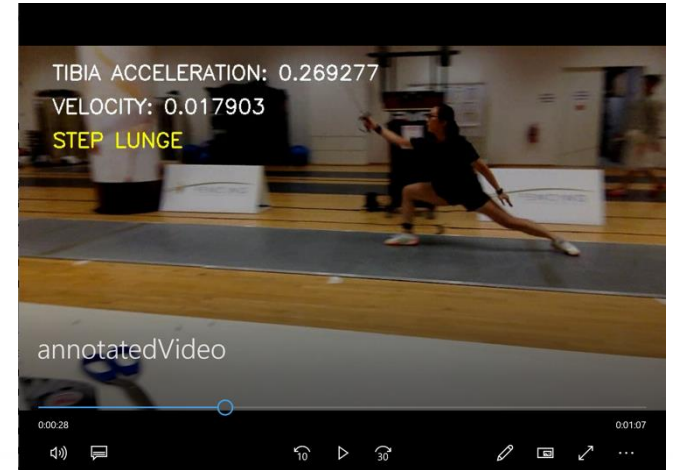
The way you walk says a lot about you



Fencing Profile Analysis System

Sensors in Sports Technology.

Fencing specific movements can be used to improve training and minimise injury.

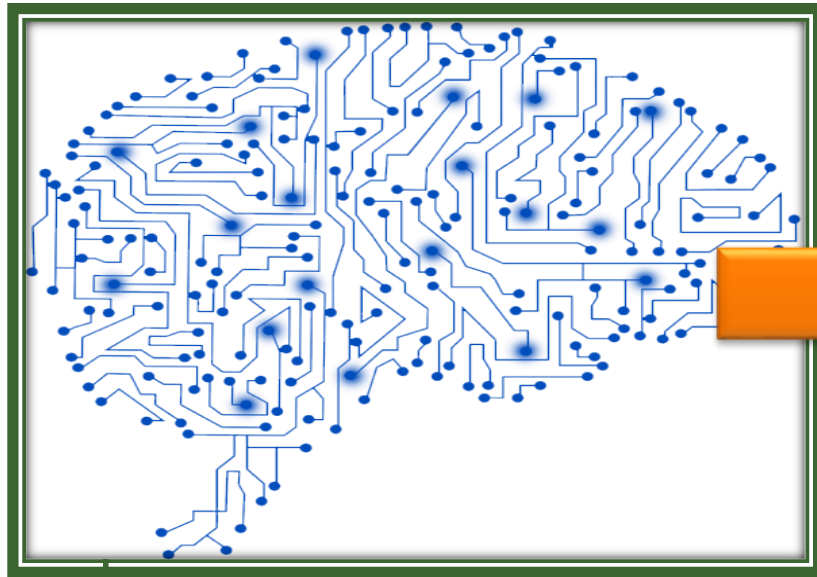




Fang Sihan

- I'm from Shanghai, China.
- I'm a third-year undergraduate majoring in Computer Engineering.
- I love sports, I play badminton and squash—and I enjoy hiking!

Course Overview



Data
Intelligence

2 Students



Mobility
Platform

1-2 Students

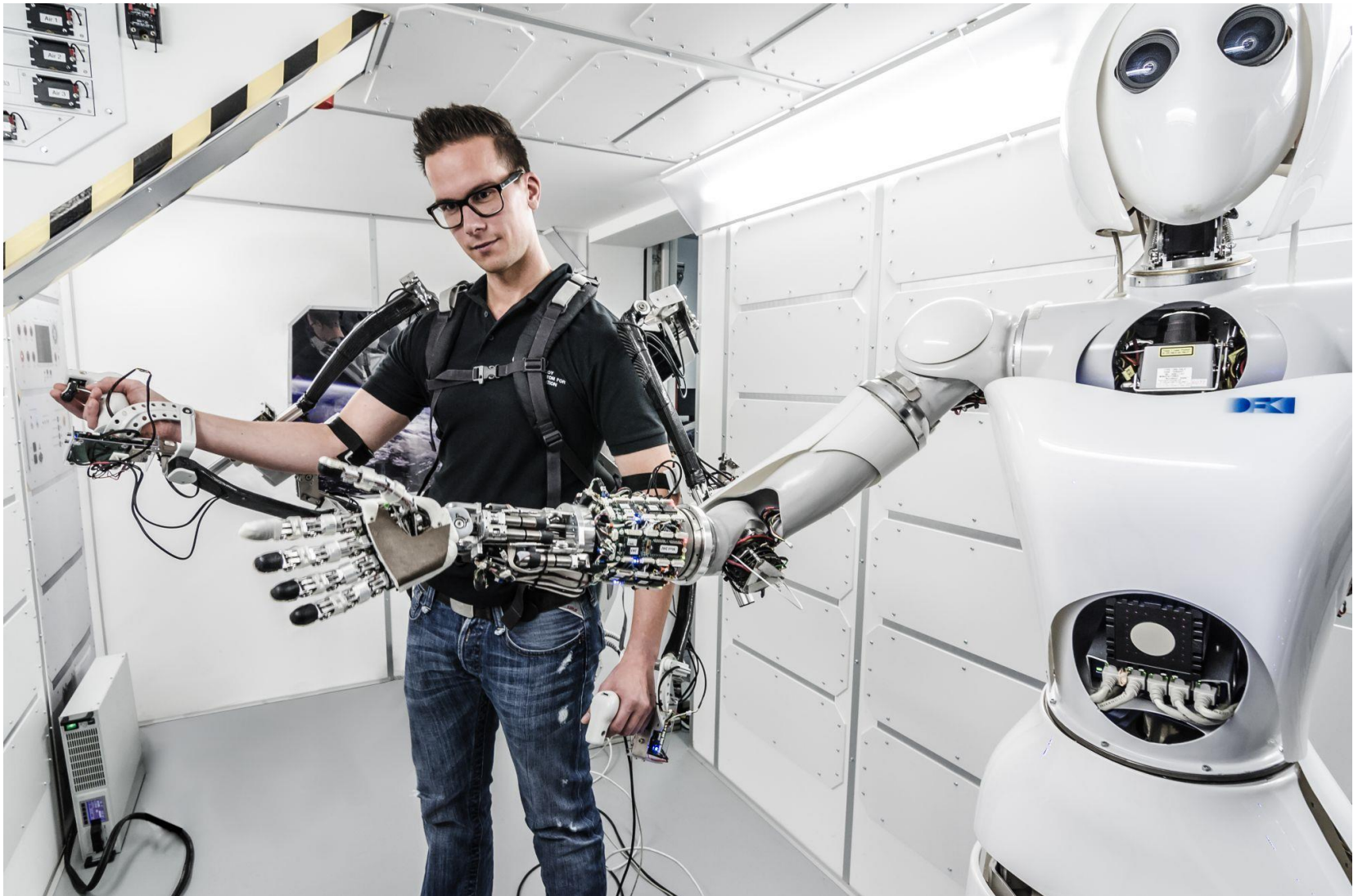
Course Objectives

- An appreciation of embedded system platforms
 - Understand the process of incremental, modular system design
 - Ability to implement a solution that impact the **physical world**
 - **Have an intensive but fun journey**
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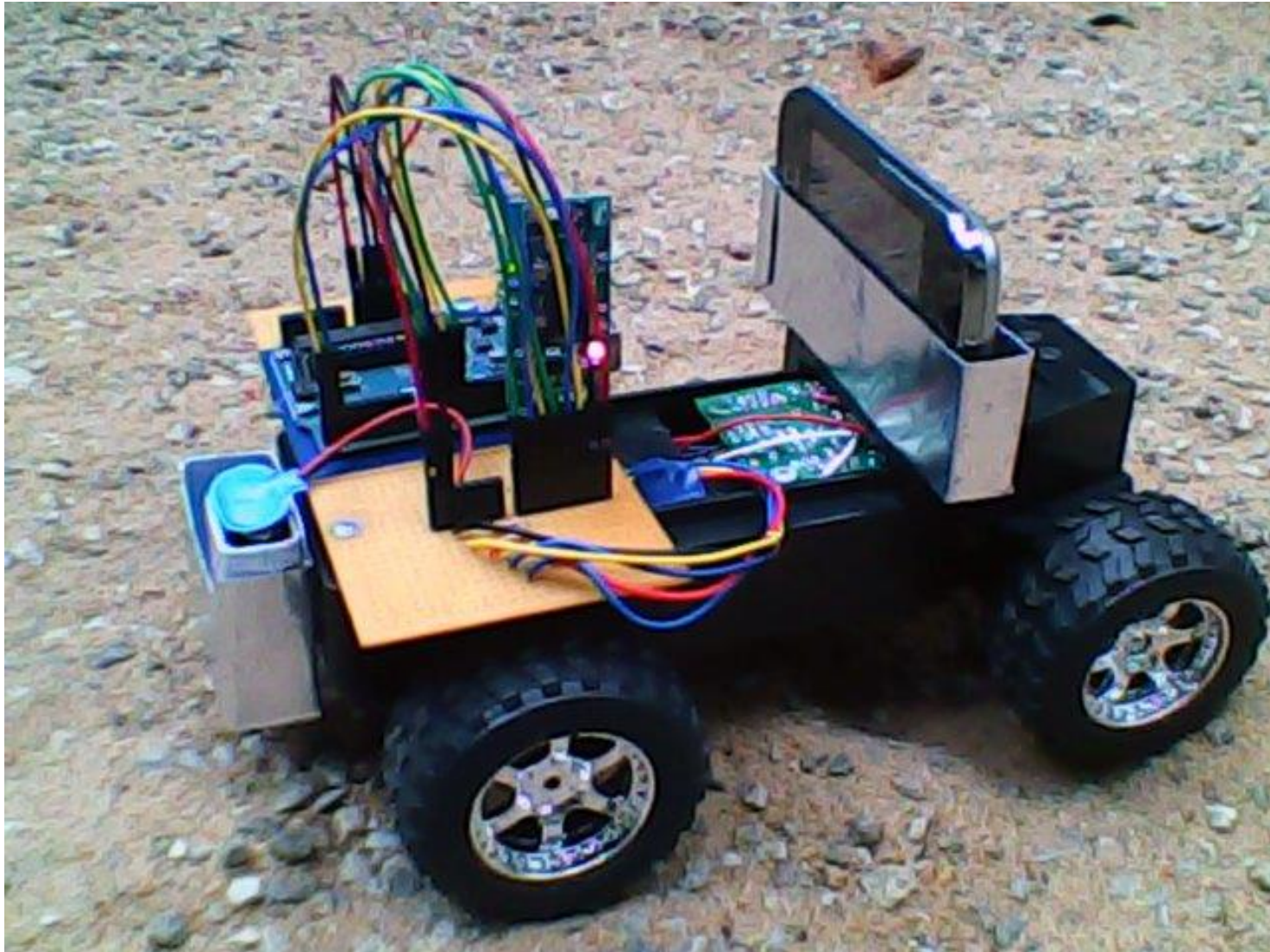
Outline

- **Robotics/TeleOperation**
 - **Platform Overview**
 - **Project Overview**
-

Requirement – Tele-Operation



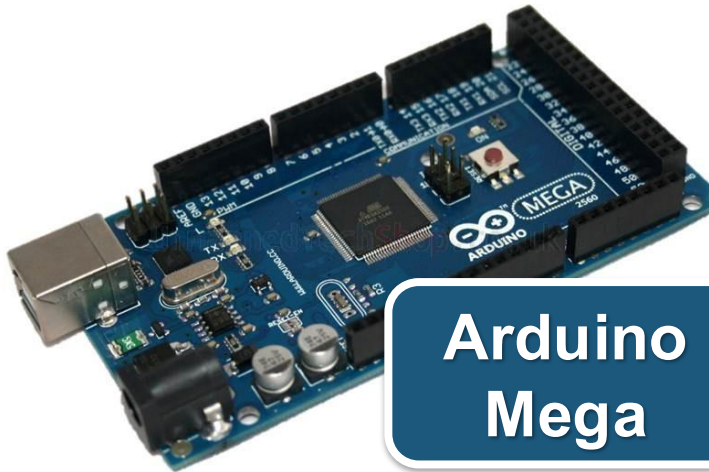
Requirement – **Mobility**



Requirement – Navigability



Hardware Platform

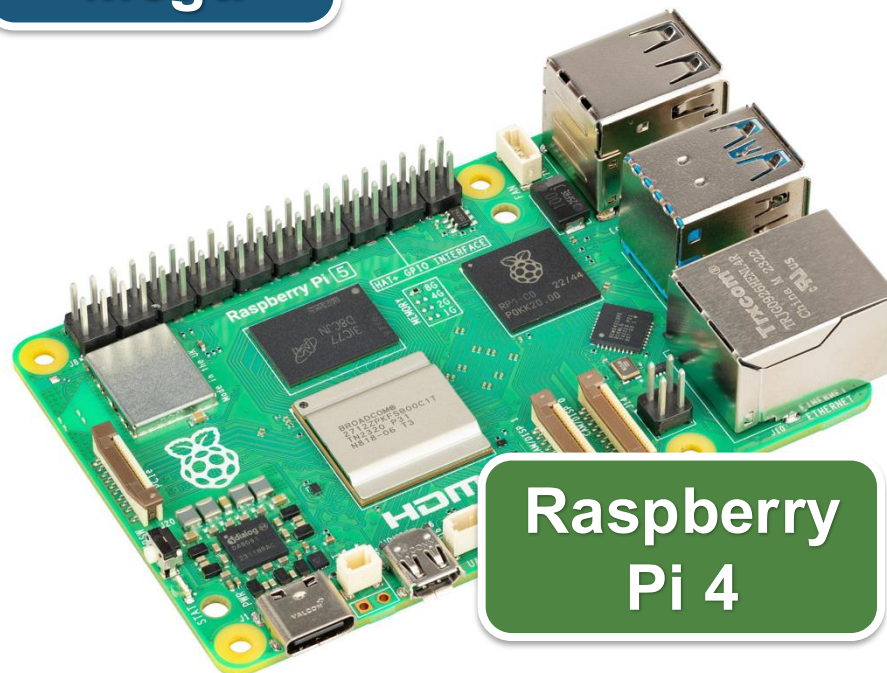


**Arduino
Mega**

16MHz
(CPU Speed)

256KB
(Code)

8KB
(Data)



**Raspberry
Pi 4**

4 x 1.2GHz
(CPU Speed)

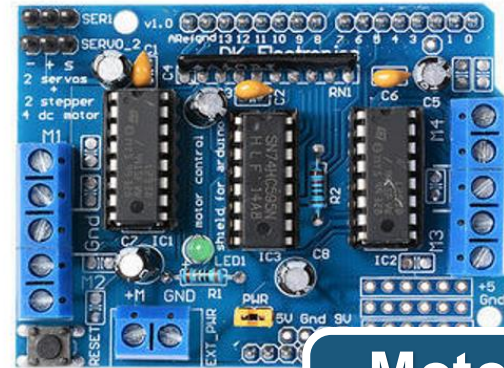
≤ 32GB(?)
(Storage)

1GB
(Memory)

External Components



Pi
Camera



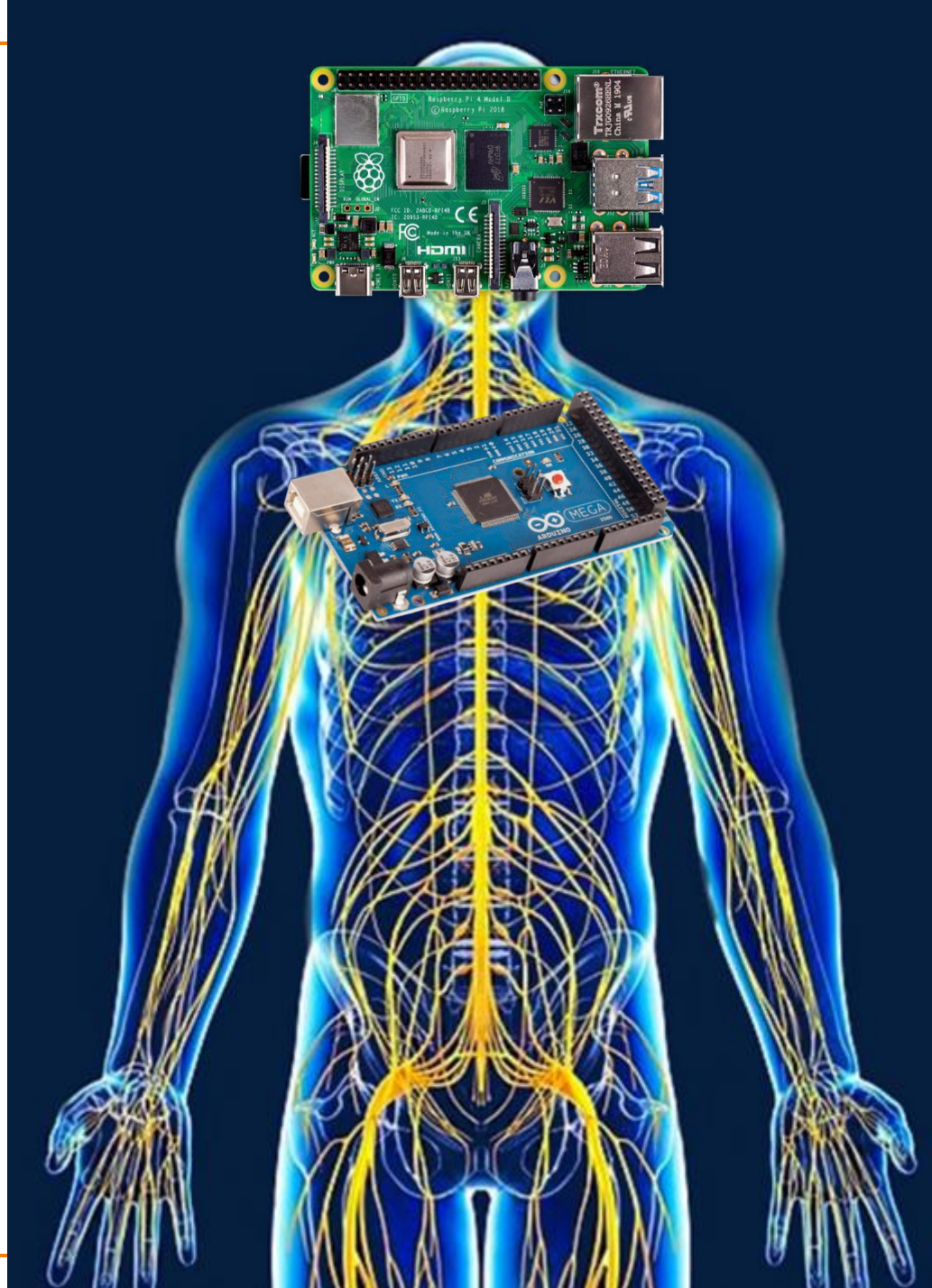
Motor
Driver



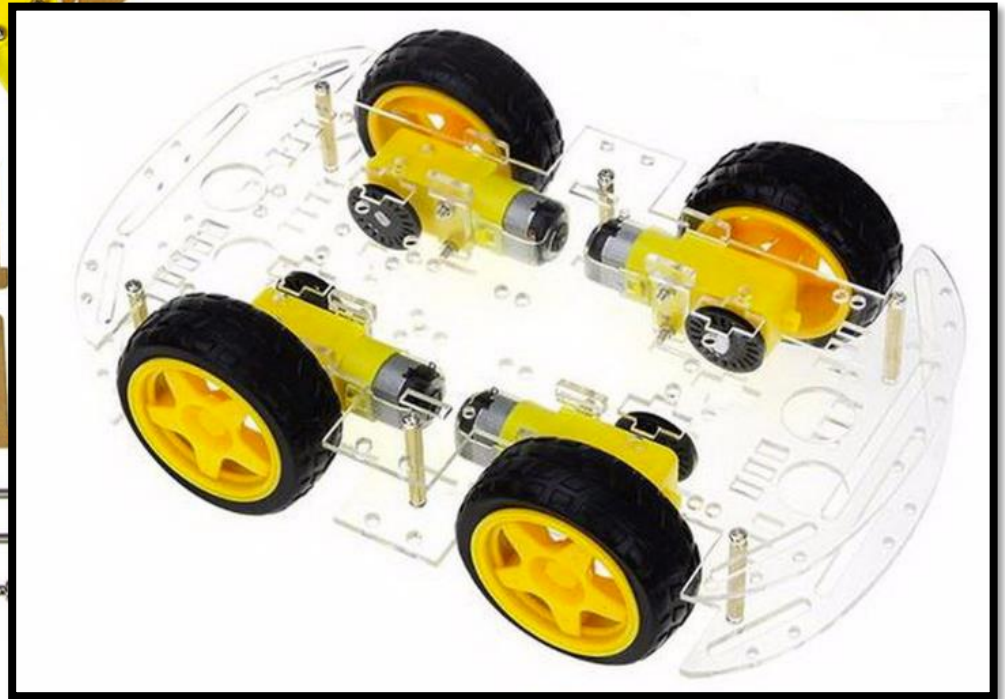
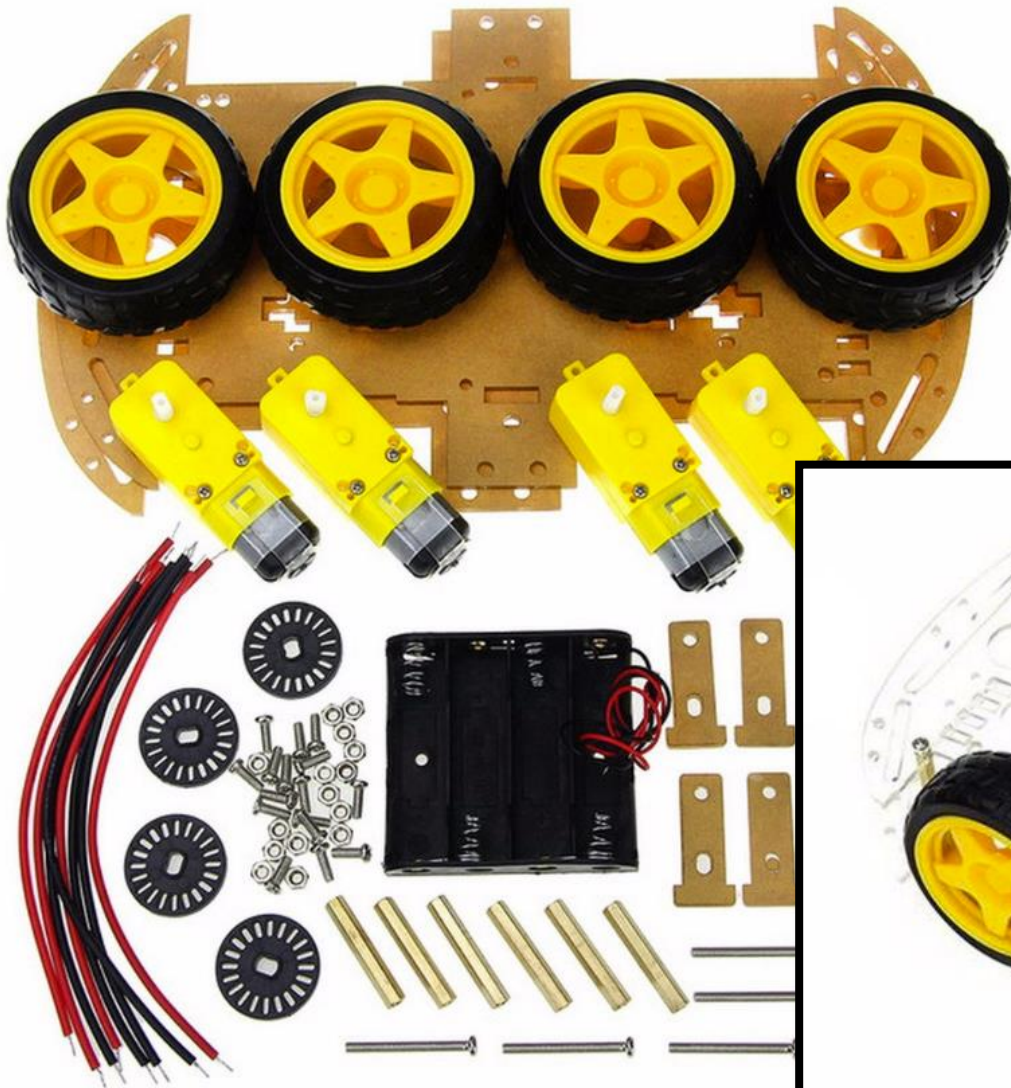
Ultrasonic
Sensor



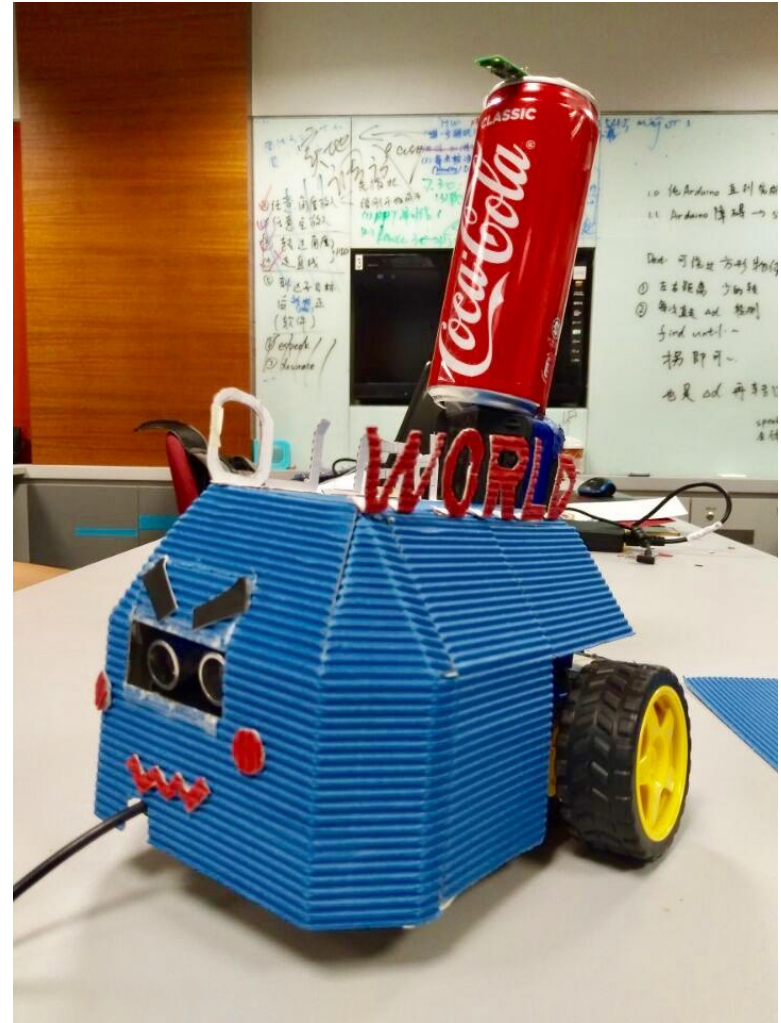
Gyroscope
/ Compass



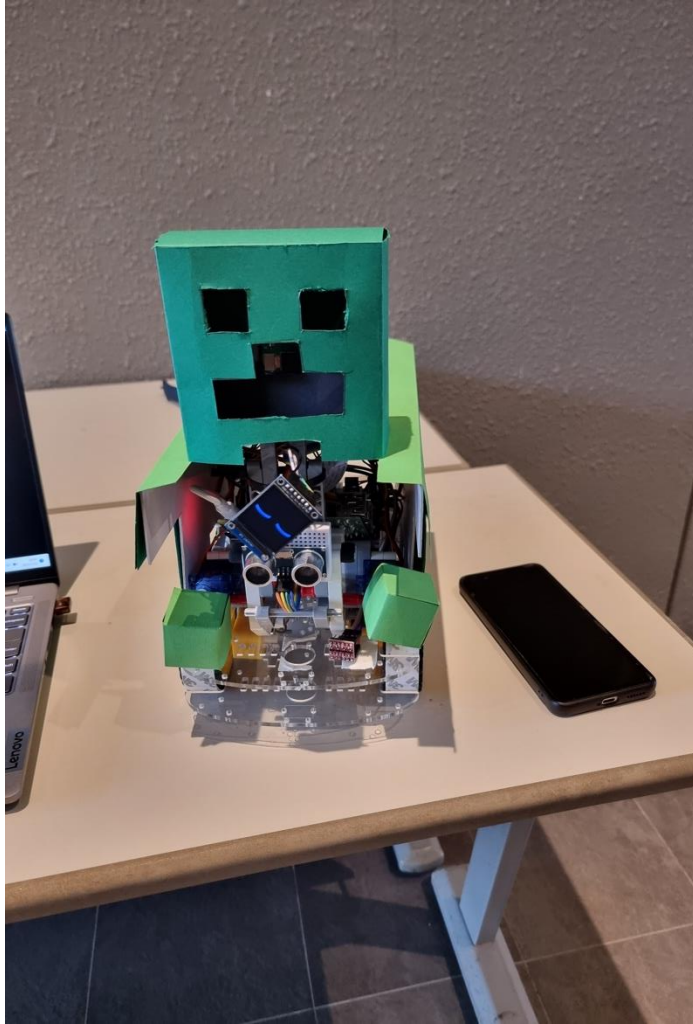
Vehicle Chassis



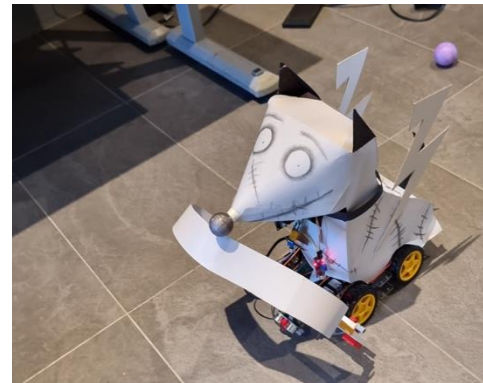
Customization – Make it yours!



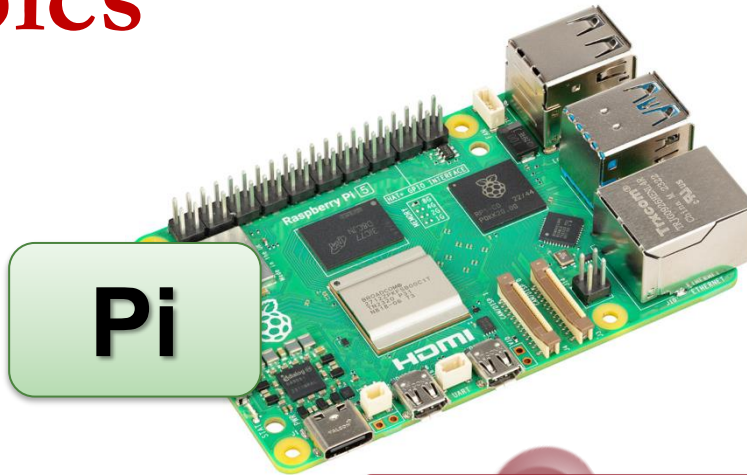
Customization – Make it yours!



Customization – Make it yours!



Topics

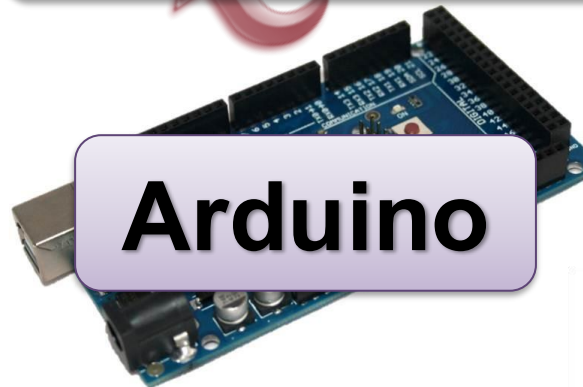


Pi

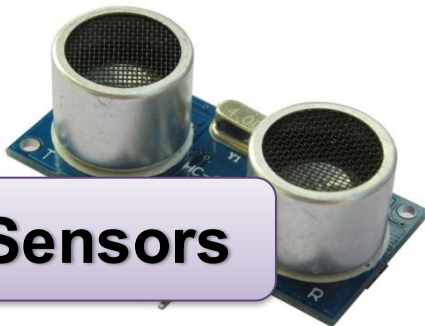


Pi Camera

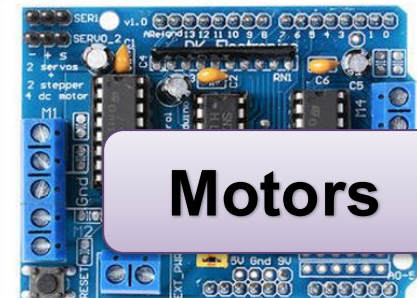
Communication



Arduino



Sensors



Motors

Project Schedule

Topic Schedule (Robotics)

Topic	Category	Day
Getting Started with Pi	Pi	1 (Thu)
Arduino Basics	Arduino	2 (Fri)
Robot Building / Chassis	Systems Integration	5 (Sat)
Robot Building / Chassis	Systems Integration	6 (Sun)
External Components	Sensors/Actuators	3 (Mon)
Serial / RTOS	Communication	4 (Tue)

Hmm... ~19 days?

D4: Integration

D5-8: Robot / Integration

D9: Treasure Hunt

D10: Advanced
Model Design

D11-16: Advanced
Model Implementation

D17: Evaluation

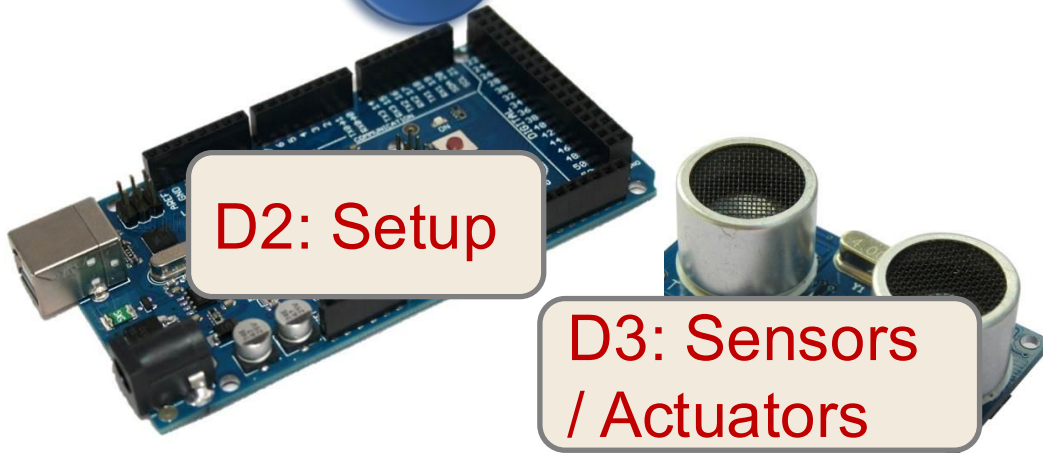
D19: Showcase

D1: Setup

D4: Communication

D2: Setup

D3: Sensors
/ Actuators



Project Overview

$$1 + 1 > 2 ?$$

■ **Baseline Model:**

- ❑ Remote controlled vehicle with camera
- ❑ Ability to classify images
- ➔ **"Treasure Hunt" Robot!**

■ **Advanced Model:**

- ❑ **Your imagination is the limit**
 - ❑ We provide components, consultation, suggestions and evaluation to **push you to the limit!**
-

Advanced Model Examples

■ **Mobility** + **Intelligence** = ???

■ Some possibilities:

❑ **Autonomous / Remote controlled explorer**

- Map out area
- Take photo of locations
- Identify / flag the locations (e.g. office occupier)

❑ **Search and rescue bot**

- Explore area to look for survivor (humanoid or otherwise)

❑ **Semi-autonomous explorer**

- Navigate via "signpost" along the path etc.

The **last slide!**

- Highlight of SWS3009:
 - ❑ You will build a **real platform** interacting with **real world**
 - ❑ Pick up practical know-how (tensor flow, electronics etc)
 - ❑ You can decide how crazy / fun the final deliverable
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Course Teaching Modes

- Learning by **doing**
 - "Learning by googling 😊"
 - Learning with **peers**
 - Group based
 - Learning by **exploration**
 - Look for alternatives, no definite answer
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Course Caveats

- The coverage is **HUGE**
 - We won't be able to cover most topics in details
 - Hardware debugging is "**messy**"
 - Expect to spend time on strange, hard to debug problems
 - Time is **SHORT**
 - Extremely condensed coverage
 - Hands-on takes time
-

END
